

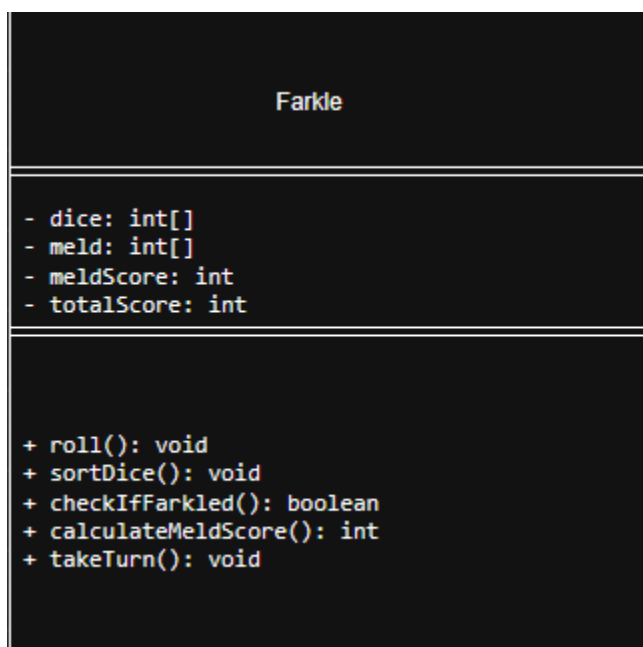
Summary:

Farkle.java's purpose is to run the very first role of a farkle game. The user should be able to add or remove dice combos into the meld and see the corresponding score of those combos. The program should automatically detect if the roll is a farkle and has no possible scoring combinations. The program should always display the highest possible meld score.

Design:

I choose a design based off the logic of the provided C++ document. I have two main arrays, one containing the hand and one containing the meld. These arrays work with each other to make sure one is always empty and one is always full for each dice. The scoring system works from the top down. It checks first for 6 dice combos and then checks for 3-5 combos before finally checking for single dice combos. All six dice will always be displayed. The combos are calculated by looking at the number of a times a specific number appears.

UML Class Diagram:



Issues:

I struggled with edge cases originally in my original build. My scoring calculations was not consistent so I switched to the number of times a specific die appears. I did this because to change my original system would have been much more complicated than making the switch. I also ran into an issue displaying the die in the original order they were generated. I

had 4 different ArrayLists that were all being constantly changed and edited. I decided to just display the dice in a sorted order.

Do Differently with more time:

I would have displayed the dice in their original order. I would also change the roll function to be based off the number of dice that have not been banked to the meld.